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Variational Auto encoders for Improved Breast Cancer Classification

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Abstract

Among women breast cancer is the second leading cause of cancer. Emergence of Artificial Intelligence(AI) in the medical care leads to good survival rate by diagnosing and effective prognosis of the breast cancer patients. Scientific findings show that supervised deep learning model is highly dependent on the size of the training set, which must be manually labeled by experienced radiologists. The freely available biomedical imaging dataset are small in size mostly. In addition, obtaining large medical image files is difficult due to privacy and legal reasons. Thus the deep learning models tend to overfit and fail to provide a general result. Also the recent studies indicate that early detection of breast cancer (BC) is crucial to achieve favorable treatment results and reduce associated mortality. Data augmentation is the most widely used approach to address the aforementioned problem. Our proposed architecture consist of the pectoral muscle removal of the dataset, then the variational auto encoders used for data augmentation and then the U-Net and its varients used for breast cancer classification. The dice similarity score obtained for the model after the pectoral muscle segmentation is 97.53% and the classification rate accuracy is 98.5% by use of the Attention Residual U-Net than the other two models.

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Keywords: Breast cancer; Data augmentation; Classification; Convolutional neural network; Auto encoders

1. Introduction

Breast cancer is a disease that develops over the breast as a result of human tumor cells proliferating. The cells infected by the breast cancer can damage the healthy cells faster which is considered one of the most dangerous forms of cancer. The lymph nodes, lungs, liver, and bones are the body parts where breast cancer(BC) is found. Breast cancer is arising at higher risk to women of all ages. The various risk factors for the breast cancer are age, family history of breast cancer, obesity, smoking, heavy alcohol consumption, radiation and postmenopausal hormone therapy. During the breast cancer screening campaigns the number of women participating has increased. Fig. 1 shows the breast cancer statistics in India, it is clearly depicted the emergence of breast cancer than other cancer types. It is necessary

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to identify the type of cancer in order to decide whether the patient needs therapy or treatment, and whether the cancer is benign or malignant. The cancer can recur after treatment but with a more fatal outcome. So the process of detecting recurrence of breast cancer after treatment is called cancer prognosis. In women with early breast cancer, the disease returns within the first five years after diagnosis and treatment[1].

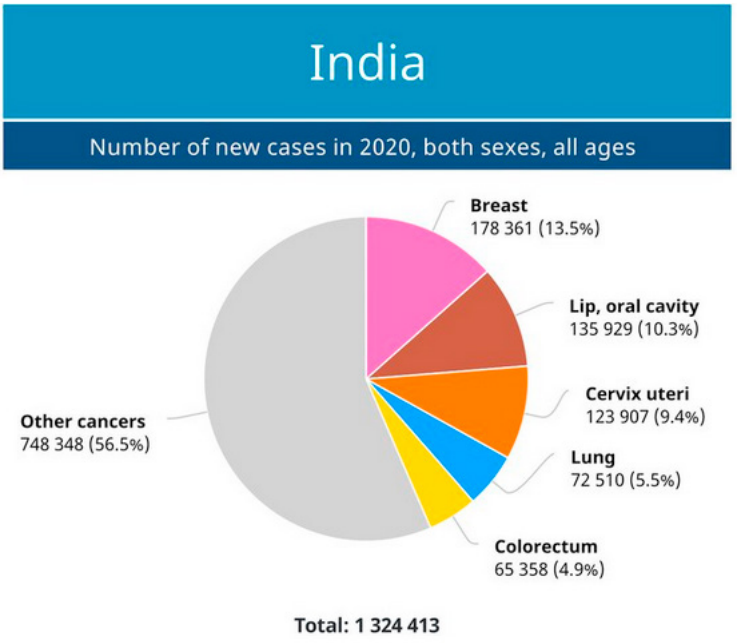


Fig. 1: Breast cancer statistics in India

Deep learning (DL) is a popular approach to preprocess and analyze images for breast cancer diagnosis. It is frequently applied to a range of computer vision problems, including segmentation and object recognition. Convolutional neural networks, or CNNs, are capable of automatically extracting features from a vast amount of data. These networks have been successfully used in biomedical image analysis for a variety of tasks, including the classification of lesions, abnormality, and suspicious regions[2]. Deep learning-based results or solutions can be used to assist radiologists in making informed decisions and planning for treatment[3]. To ensure that a DL model is able to infer knowledge from data, and to avoid overfitting, it is important to have a high level of ground truth in order to obtain a large dataset of high quality and annotated data.

Data Augmentation method will help in generating new images with some transformations on the orginal image. So that the overfitting problem as discussed above can be rectified. This augmented data can be used to characterize the collection of input observations and minimize the gap between the training and validation sets. There are other ways to lessen the impact of overfitting and class imbalance besides data augmentation. The other techniques that can be used in DL models to avoid overfitting includes batch normalization, dropout, transfer learning, pre-training and early stopping. The remainder of this paper is discussed here: section 2 describes the related work, in section 3 the proposed architecture is explained, the validation result is explained in the section 4 and finally the conclusion and future scope.

2. Related works

Data Augmentation is a technique used to expand the size of a training dataset to accommodate larger-scale learners. It involves introducing random variations into existing training samples. Such as those from [4] by making various modifications to the original images to increase the dataset size. However, this technique does not significantly

improve the model training or generalisation of test data with correlated images.By using the flipping technique, the original image is transformed into a mirror image with both vertical and horizontal axes, with horizontal axis flipping being preferred over vertical axis flippingThrough the use of an axis that rotates the images left or right within the range [1°, 359°], the rotation method converts each image into a unique translation vector. Rather than focusing on traits that are unique to one spatial region, this method helps the network learn geographic invariant properties[5].

According to[6], the best algorithm for detecting breast cancer is the Random Forest (RF) method. Researchers studying machine learning have recently turned their attention to the ensemble of ML algorithms in an effort to get over the drawbacks of one method and improve the accuracy of BC treatment. To determine how successfully contemporary deep-learning algorithms can diagnose breast cancer, seven deep-learning architectures are being put to the test. The Long Short Term Memory and Gated Recurrent Unit (GRU) designs in this study surpassed every other method for BC diagnosis.

In a variety of studies, the use of additional breast cancer imaging data, such as breast histopathology pictures, to support breast cancer categorization was investigated. A model utilizing Generative Adversarial Network to improve images of the histology of the minority class [7]. The class imbalance and lack of data issues can be greatly resolved by using deep models based on attention mechanisms. A deep learning model's architecture can also be changed to increase computation complexity and learning time, for example, by fusing deep learning with wavelet networks.

A deep segmentation method called Conditional Generative Adversarial Networks (cGANs)[8] consists of a generator network that has both an encoder and a decoder. While the decoder layer uses the information to create a binary mask, the encoder layer extracts properties like texture, edge, and intensity. A mask for mammography masses is created by the generator network and sent to the classifier. An attention gate (AG) based densely connected U-Net breast mass segmentation technique. The decoder is combined with AGs to improve the segmentation of the U-Net model, and the encoder is firmly coupled to CNN to operate as a feature extractor of the various shapes and sizes of breast masses. The process of automatically segmenting cells in bright field microscopy is difficult because of the presence of image artifacts, low contrast, overlapping cells, and a large range of cell variations. Moreover, the scarcity of labeled bright-field images further hampers the capacity to develop supervised models for automated cell segmentation. This paper presents a novel cell segmentation framework called Saliency and Ballness driven U-shaped Network (SBU-net) to tackle these challenges [9]. The design incorporates a novel data-driven feature fusion module that enhances the discernible structure of cells by leveraging their saliency and roundness characteristics. The utilization of an encoder-decoder model that incorporates dilated convolutions, together with a distinctive combination loss function, successfully captured the comprehensive context of cell topologies and produced accurate cell segmentation results.

The K-nearest neigbour and Parzen Window algorithms are commonly employed in medical diagnosis and disease classification as generative algorithms. The healthcare data created or gathered is characterized by a high number of dimensions and cannot be presumed to conform to a specific distribution. The usual methods are ineffective because they are hindered by the high level of computational complexity, the curse of dimensionality and the presence of changing distributions. The [10] focuses on a fusion technique to assess closest neighbor algorithms using several parameters, including data size, data dimensions, window size, and number of nearest neighbors. A novel method for classifying high-dimensional data by utilizing a depth representation of subspace clusters. Approach involves dynamically selecting the nearest neighbors for classifying a query point. This is done by analyzing the data's structures and distributions, extracting relevant features, and determining the optimal value of k based on the data's structure using a data depth function. In this study[11] was to assess the long-term oncological prognosis for women who received curative breast cancer treatment. This study is a retrospective cohort analysis of 1301 patients with breast cancer, spanning all stages, who underwent primary treatment at a single Indian cancer facility with the goal of curing their disease between 2004 and 2010.The study utilizes authentic healthcare records data obtained from hospitals operated by HealthCare Global Enterprises Ltd[12]. The dataset is analyzed with respect to its four main class variables: mortality, progression, recurrence, and metastasis.

It is recommended to use wrapper feature selection processes in addition to the final ensemble model in cases where the filter approach cannot produce a specific group of features [13]. Also the imprecise method of estimating the density of benign or cancerous tissue. Cancer data sets contain a multitude of patient characteristics, not all of which are useful for cancer prediction. In these cases, feature selection techniques [14] are useful to maintain the appropriate feature set. A deep learning methodology is employed [15] to classify histopathology images of breast cancer. A

hybrid fusion of Inspection-ResNetv2 and EfficientNetV2-S, utilizing pretrained weights from ImageNet was used. The suggested model underwent validation using the BreakHis and Breast Cancer Histology (BACH) dataset. For every class, the impact of the same 11 predictor factors is investigated. The fundamental concept is to transform the mammography image into a three-dimensional matrix[16]. The obtained matrix is utilized to create a binary image from the mammography. Numerous methods have been employed, including cell detection, border removal, object smoothing, structure detection and huge object extraction. Ultimately determining the thickness of tissues in an image without having to segment every object separately. An overview of the most prevalent types of breast cancer[17], their staging, and the various methods and techniques for diagnosis. This survey offers information on all breast cancer detection modalities and methodologies, compares the cost and accuracy of each and offers insights into the utility and efficacy of each methodology in relation to the type and staging of breast cancer.

3. Materials and Methods

This section briefly explains the proposed architecture and individual operations carried out in the study. The mammogram images consists of pectoral muscles which may lead to misclassification as malignant one, inorder to mitigate this issue in the proposed model the canny edge detection and hough transform is implemented in the preprocesing phase. Then there is a need for data augmentaion due to the scarcity of images in the dataset, this limitation is achieved by use VAE for the data augmentation technique. These are the advantages of our proposed model. The dataset is first preprocessed by use of the median filter and then the pectoral muscle of the mammogram image is carried out by use of the Hough transformation. And then the synthetic generation of preprocessed breast cancer data for the classification task is carried out. The generation or augmentation of the dataset is carried out by use of the general geometric transformations and variational autoencoder (VAE). Fig. 2 depicts the study's overall research flow. The suggested architecture, as shown in Fig. 2 consist of input as the mammogram images, which is being pre-processed by median filter, CLAHE and the pectoral muscle removal of the image is carried out using the Hough transform. The pre-processed images are augmented using the general geometric transformation and VAE. On this pre-processed and augmented images the breast cancer classification process is carried out. Classification of the breast cancer images is carried out by use of the U-Net, Attention U-Net and the Attention Residual U-Net.

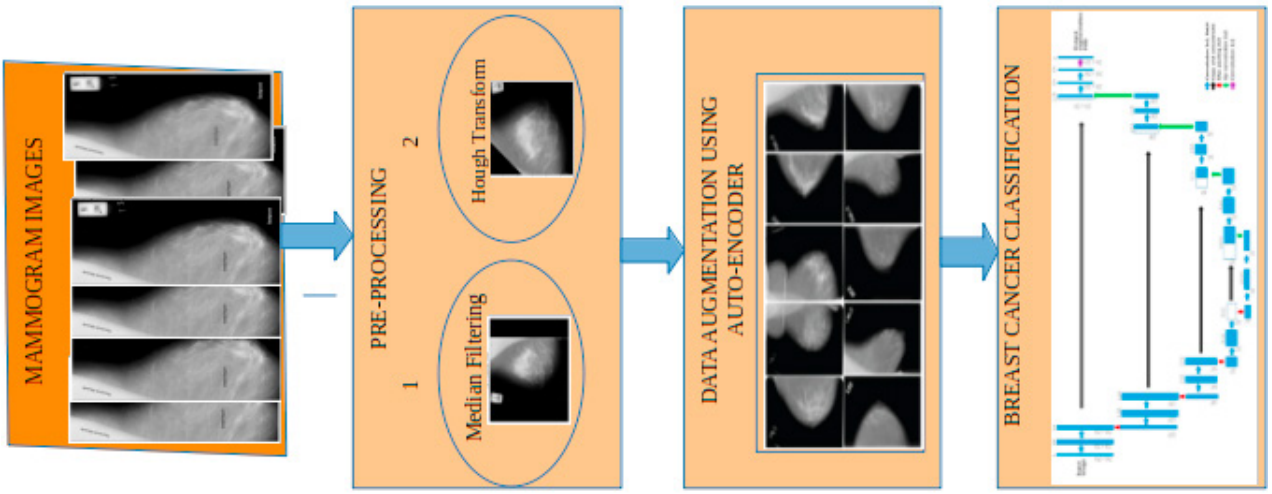


Fig. 2: Proposed Architecture

3.1. Dataset

Mammography images from the MIAS data set were used in this study [18]. To better understand and track female breast cancer, a group of enthusiastic researchers built the MIAS dataset. 312 pictures, comprising both left and right breast photos, from 161 patients are included in the MIAS dataset. The dataset consists of mammography images that radiologists have manually identified. The MIAS dataset's mammography images were initially 1024x1024 pixels in size. The original dataset consists of 332 mammography pictures in grayscale. There are six categories of anomalies that are characterized based on their severity as either benign (B) or malignant (M). Scans that did not show any abnormalities were assigned the value (A) for classification purposes without any specific criteria. The images are categorized as sets of films, where each set corresponds to the left (even filename values) and right mammograms (odd filename values) of a single patient. The Mammographic Image Analysis Society (MIAS) is a consortium of research organizations in the United Kingdom that specialize in the scrutiny and evaluation of mammograms. MIAS has curated a collection of digital mammograms for the aim of doing research. The UK National Breast Screening Programme has digitized its films using a Joyce-Loebl scanning microdensitometer. This device quantifies the optical density of individual pixels and expresses it using an 8-bit binary code. The pixel resolution is 50 micrometers. The University of Essex's Pilot European Image Processing Archive (PEIPA) offers access to mammographic images.

3.2. Preprocessing

In image dealing with the filtering is to smoother either the high recurrence or low recurrence in an image. It is utilized to recognize either edges in the image and smoothing or redesigning the image. For pre-processing mammogram images exceptional kinds of filtering frameworks are accessible. Filtering is performed in the spatial region. The fundamental filtering strategies utilized in the proposed work are: Median filtering, Mean filtering, Adaptive Median filter and Wiener filtering. In this paper the exhibition of the denoising filters(mean, median, wiener and adaptive median filter) conveyed on the quality boundaries like MSE,SNR and PSNR. The sharp edges is better saved and is less fragile to anomalies in median filters than mean filters.

Pectoral muscle have intensity region on mammograms having no huge data about tumor acknowledgment and may cause mistaken pathological decisions. Along these lines, expulsion of pectoral muscle from breast parenchyma is thought of significant properties of computer aided diagnoses framework for precise tumor segmentation and classification. In this proposed work, filtered mammograms are fed to the canny edge detector for detecting the boundaries of the breast region from the pectoral muscle. After obtaining the boundary region, the hough lines are marked over the boundary and then pectoral muscle is removed from the mammogram. The proposed algorithm for pre-processing is shown in Algorithm The steps included in the algorithm are: Collect mammogram images, pre-processing using median filter, pectoral muscle segmented, applying Canny edge detector on the pre-processed image, applying Hough transform on the edge detected image, removing the pectoral muscle detected using the Hough transform and finally displaying all the images.

In this section the pectoral muscles are segmented out from the mammogram images by use of this canny edge detection and hough transformation as explained the Algorithm 1. The pectoral muscles needs to be segmented out from the mammogram images to avoid the mis-classification of tumor predction.

3.3. Data Augmentation

The artificial information created from discriminator and unknown noise attempt to discern between false and real data for the Generative Adversarial Network (GAN). Based on information from the discriminator, the nature of data that has been generated artificially is improved. A few researchers have utilized GAN to produce artificial clinical imaging data. A GAN's generator begins with noise that is random and attempts to adjust its loss depending on the discriminator's loss, also GAN-based designs are inherently unsupervised. The categorical feature sections in the clinical datasets are unbalanced and of a highly sensitive character. Data augmentation makes the model more broadly applicable. It reduces overfitting of the data and increases data variability. The expense of gathering and classifying more data is reduced. It raises the prediction accuracy of the deep learning model. Fig. 3 illustrates the graph of models with or without overfitting, with the expected result appearing on the left side of the graph. The overfitting can be avoided by use of data augmentation techniques to increase the count images in the dataset.

Algorithm 1: Proposed Algorithm

Input: Mammogram images
Output: Binary classification
Data: MIAS dataset
Def `MedianFilter(`*sourceimage, kernelsize*`):`
  Removing noise from the source image by use of kernal size
  return medianfilteredimage
Def `Cannyedge(`*medianfilteredimage*`):`
  Detecting edges of the breast
  return cannyimage
Def `HoughTransform(`*cannyimage*`):`
  Marking hough lines on the detected edges
  return houghlines
Def `Removal(`*Houghlines*`):`
  Removing pectoral muscle region from the mammogram images.
  return removedimage
Def `VAE(`*removedimage*`):`
  Augmenting using the VAE.
  return augmentedimage
Def `Classification(`*augmentedimage*`):`
  Classification of the augmented images using the U-Net, Attention U-Net and Attention Residual U-Net.
  return classifiedoutput
Def `Display(`*filename*`):`
  medianfilteredimage = MedianFilter(source image, kernel size)
  cannyimage = Cannyedge(medianfilteredimage)
  houghlines = HoughTransform(cannyimage)
  removedimage = Removal(houghlines)
  augmentedimage = VAE(removedimage)
  classifiedoutput = Classification(augmentedimage)
  Binary classification of the mammogram image is carried out.
  return 0
Function `Main`**:**
  image = display(filename)
  return 0

It is possible to use contemporary deep generative architectures known as variational autoencoders in generating new data that has been synthetically derived from historical data. A variational autoencoder based model consists of two segments: an encoder and a decoder. The neural network encoder and decoder sets are used to compress real-world input in order to comprehend the underlying probability distributions and generate new instances based on the inferred latent space probabilities. The taught probability distribution, which the decoder has learnt from the encoder's latent probability distribution, is generated or mimicked by the decoder to reconstruct the original input. In order to combine various separate models, the Hierarchical Variational Autoencoder (HVAE) goes beyond the usual meta-learning techniques. The VAE architecture as shown in Fig. 4 whose input is the mammogram image being fed into the encoder where the hidden layer and the embedding is represented. The embedding includes the mean and standard deviation which make up each Gaussian distribution latent component. The VAE generative embedding consists of all Gaussian combinations. The data augmentation is achieved by use of this VAE where the mammogram image is fed into the encoder unit. The mean and standard deviation is calculated and from which the latent space or the features

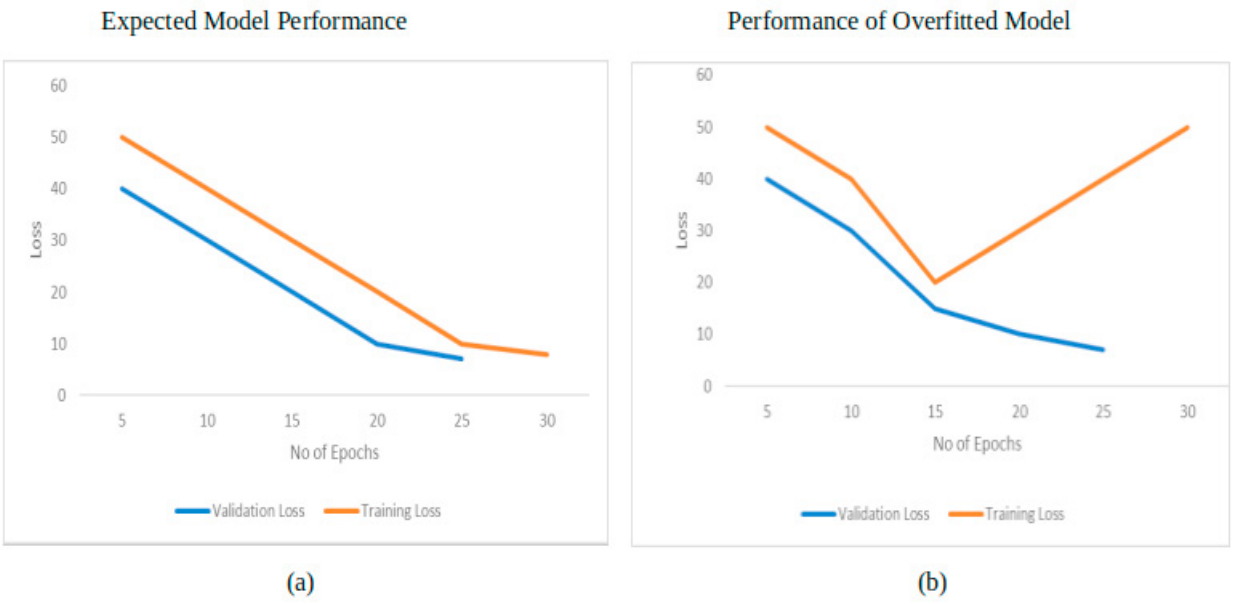


Fig. 3: (a) The error functions for training and validation are dropping. (b) The undesirable outcome of overfitting, which is a reduction in the training error

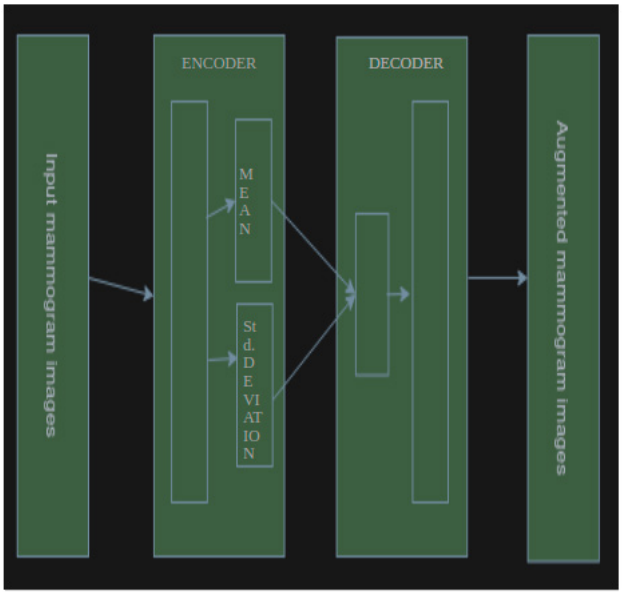


Fig. 4: Variational Auto-Encoder(VAE)

are extracted and then fed into the decoder unit for regenerating the images. This way the augmented images are obtained by use of this VAE technique.

A similar architecture is utilized for each low-level VAE, however on a fundamental level, these could be freely planned and further refined for a particular information source and job needing to be done. HVAE operates on a per-network basis, it is intended to address some of the shortcomings of X-VAE and MM-VAE. Specifically, representations acquired from a few distinct low-level autoencoders make up the contribution to the high-level

autoencoder. As a result, the contribution to the high level autoencoder currently consists of approximated multivariate standard normal distributions characterizing the overall attributes of the individual input modalities. These low-level autoencoders currently perform distribution regularization terms in each of them independently. Furthermore, because each information source is handled modularly, HVAEs are appropriate for handling information sources that use concentrated low-level autoencoders. Building an HVAE involves a two-stage approach to learning where minimal VAEs need be created first and the final high-level representation can be learnt based on the output of the low-level encoders, building an HVAE adds a large processing burden compared to the other three structures.

Adequate data is necessary for effective training of neural networks else the overfitting may occur in the model. When dealing with a low data scheme, the parameters are insufficiently specified, resulting in inadequate generalization of the networks. Data augmentation strategies improve the generalizability of the deep learning model by effectively utilizing the original data. Conventional data augmentation methods, such as random translations, flips, rotations, and the introduction of Gaussian noise, produce genuine but limited amounts of data. Variational Autoencoders have been employed for the purpose of generating synthetic samples and enhancing the performance of classifiers. The generation of artificial data for underrepresented classes might be advantageous for several purposes. One primary motivation for generating synthetic samples is to circumvent the need to use real samples due to privacy concerns. There is a significant risk of misuse or unauthorized access when dealing directly with the database that contains the original samples. Another factor is the excessive sampling of the minority classes. In practical situations, there is always a chance that certain classes may be inadequately represented.

3.4. Classification

Motivated by the results of UNet and its diversity to advance the overall presentation. suggested a design for bridging two basic UNets. The original UNet design concept includes the skip associations between an encoder and a decoder network and then make a further adjustment of skip associations in the other direction between a decoder and an encoder network before establishing a second UNet. Afterwards, two flowing encoders and decoders are shown in the final design, which are all connected by different skip associations. The concept of recovering the fine-grained details that are lost in the UNet encoding process. By linking it to the newly decoded highlights, it is used to encode the high-resolution characteristics. The conventional UNet design is also given the Atrous Spatial Pyramid Pooling (ASPP) component and the suggested architecture is used in two separate forms, Attention U-Net (AUNet) and Attention Residual U-Net. The models is put for categorizing the breast mass that was recently found in mammograms from the MIAS dataset to use.

4. Validation Results

The MIAS informative index used in this publication was developed by a group of researchers who were interested in learning about and identifying breast cancer in women. The MIAS dataset includes 312 images of the right and left breasts from 161 patients. The radiologists' chosen mammography images are included in the collection. For each given dataset, the quality bounds of PSNR, SNR, and MSE are determined in order to assess the performance evaluation of denoising filters such as mean, median, adaptive, and wiene error in mammography images. Image pressure quality is analyzed using the MSE and PSNR. While PSNR treats a fraction of the peak error, MSE handles the overall squared error between the compacted and the first picture. The error decreases as the MSE value decreases. To process the PSNR, the accompanying calculation to compute the mean-squared error before processing the PSNR:

$$MSE = \frac{\sum_{M,N}[I_1(m,n) - I_2(m,n)]^2}{M * N} \tag{1}$$

The row and column counts of the input images are denoted by M and N in the preceding equation. Next, the block uses the following equation to calculate the PSNR:

$$PSNR = 10log_{10}\frac{R^2}{MSE} \tag{2}$$

The propose work consist of a novel approach to remove the pectoral muscle in terms of both the mammogram images using canny edge detector and hough transform. This is used to detect the edges boundaries and to extract

the mammogram image without any artifacts in it. Before refinement is achieved some pre-processing technique is used. This approach is applied on dataset furnished with the aid of using MIAS. This median filtered image is used for excision of the pectoral muscle using the canny edge detection for detecting the boundary between the breast region and pectoral muscle. After the boundary is detected hough lines are plotted over it and then the pectoral muscle region is removed from the mammogram.

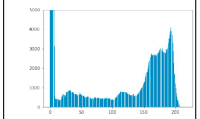
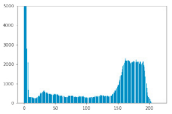
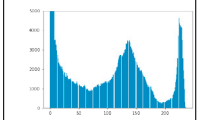
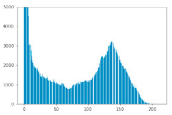
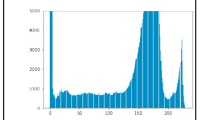
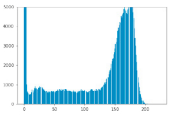
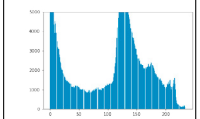
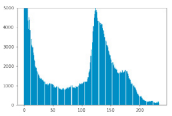
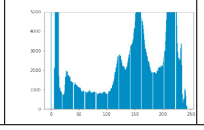
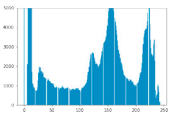
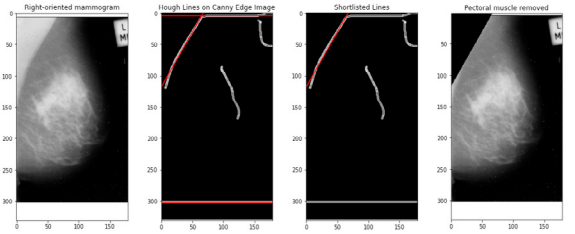
Image Id	Original image	Pectoral muscle removed
mdb018		
mdb048		
mdb157		
mdb210		
mdb320		

Table 1: Comparing the histograms of original and pre-processed mammogram images

Fig. 5: Result obtained after the preprocessing and pectoral muscle removed images



In Fig. 5, first figure is the input image after preprocessing, the second image is hough line on the canny edges, third one is the shortlisted lines after obtaining the hough lines on second image and finally the excision of the pectoral muscle i.e the final result. The histogram of the original mammogram image and the pectoral muscle removed preprocessed image is shown in TABLE. 1. In this table randomly choosing the five samples such as mdb018, mdb048, mdb157, mdb210 and mdb320 images. After comparing the histograms it is found that there is a change in histogram intensity after removing the pectoral muscle region from the original mammogram images. This will improve the feature extraction technique for the future work.

This study uses the Dice Similarity Coefficient (DSC) to assess the precision of muscle and breast border extractions. The breast and muscular border were obtained from the input image and the boundary was extracted out of the analysed image. Consequently, the overlapping area between them was determined. The DSC (Dice Similarity Coefficient) of all the images after pectoral muscle removal is calculated and obtained good results for it. The dice coefficient is calculated between two binary pictures with extracted borders. The Dice coefficient ranges from 0 to 1, with a value of 1 indicating a strong similarity between images. The Table 2 presents a comparison of the dice scores achieved by different researchers. The analysis reveals that the suggested methodology exhibits a superior DSC score.

The augmented images after the segmentation process make use of the augmentation technique a good value of dataset obtained for performing the classification task. In the Table 3 the comparison of the model without data augmentation, with normal flipping and rotation and then the proposed model data augmentation using the VAE is shown. It is clearly depicted in the table how much influence does data augmentation achieved. The techniques compared includes the original dataset without any data augmentation applied is trained in the model and got an accuracy of 92.16%, secondly applied the normal flipping and rotation of the dataset and evaluated the model over the dataset and got an accuracy of 94.55%. Finally the augmentation using the VAE is applied over the dataset and acheived an accuracy of 97.53%.

Author	DSC
Ahmad Raza et al. (2020)[19]	96.16%
Zwiggelaar, Reyer et al. (2019)[20]	97.5%
Tzikopoulos, S.D. et al. (2011)[21]	94.5%
Proposed method	97.53%

Table 2: Dice Similarity Coefficient(DSC) comparison after the pectoral muscle removal

Technique	Accuracy
Without VAE	92.16%
With normal augmentation	94.55%
Proposed method	97.53%

Table 3: Influence of data augmentation on the pectoral muscle segmented data for classification

Model	Accuracy before VAE	Accuracy after VAE
U-Net	96.16%	96.8%
Attention U-Net	97.1%	97.6%
Attention Residual U-Net	98%	98.5%

Table 4: Classification rate of U-Net and its varients before and after data augmentation using VAE

After the application of VAE for data augmentation the accuracies obtained by use of the U-Net and its varients is depicted in Table 4. It is found that the Attention Residual U-Net gives good classification accuracy than the other models.

5. Conclusion and Future Work

In the modern era, the intelligent analysis and categorization of clinical images has been an emerging topic. We have suggested an alternative arrangement for the sequence of mammography images of breast nodules in this particular case. The suggested method combines elements of two different approaches. U-Net, Attention U-Net, and Attention Residual U-Net are employed for the classification job, whereas the pectoral muscle segmentation is used for nodule segmentation. By doing this, unnecessary features from a mammography image will be removed. Deep learning is bringing with it ever-more-refined architectures to extract hidden features from a wide range of computer vision tasks. The main goal is to investigate directly how data augmentation processes affect deep learning methods for tasks like classification and suspicious region identification. Moreover, the experiment is carried out with the publically available breast cancer dataset Mammogram Image Analysis(MIAS). The dice similarity score obtained for our model after the pectoral muscle segmentation is 97.53% and the classification accuracy is 98.5% by use of the Attention Residual U-Net. In the proposed model only the variational auto encoder is used for data augmentation which has limitations while processing the encoding and decoding which result in the loss of information, leading to the production of blurry and low-resolution samples that lack the sharpness and level of detail seen in the original data. Future work will investigate how to greatly increase breast cancer classification scores by developing augmentation techniques for generating synthetic breast cancer data into tables using generative models.

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